

CSE423

Midterm Assignment

To answer the question, you will need four variables **A, B, C and D** which are sequentially the **first, second, third and fourth pair** of digits from the left in your **student ID**. For example, if your ID is **15101208**, then **A= 15, B= 10, C= 12 and D= 8**.

[Incorrect Calculation of such pairs will result in a -50% penalty in the assignment mark.]

1. Find **3D** vectors V_1' and V_2' perpendicular to the vector $V_1(A, B, C)$ and $V_2(B, C, D)$ respectively and show whether V_1' and V_2' are perpendicular to each other or not.
2. Consider a line having endpoints from **(D, C)** to **(B, A)**. Find out all or **first 10** pixels (whichever is less) of the line using the **midpoint line drawing algorithm**.
3. Consider a rectangular clip region defined by the points **(-100, min(A, B))** and **(50, max(A, B))**. For the given clip region, find out whether the line having endpoints **(0, 423)** and **(C, D)** is completely inside/outside or clipped **only from the outcodes of Cohen Sutherland Algorithm**. (that means you don't need to go to the third step, trivial acceptance/rejection is enough). Then verify your claim using **Cyrus Beck Line Clipping Algorithm** and if the line is **clipped**, find out the **clipped coordinates** of the line.
4. A composite transformation is defined by a **90-degree clockwise rotation** about point **(A, B)**, followed by a **shearing along both axis** by a shearing factor of **(C, D)** about point **(B, 0)**, followed by a **uniform scaling** by factor **C**, followed by a **reflection about the line $Ax + By + C = 0$** and finally followed by a **translation of (B, D)**. If a point **P** is transformed into **P'** with the above transformation, write down the **composite transformation matrix representation for transforming P to P' and P' to P respectively** in the correct sequence of matrix multiplications.
5. Suppose a line segment starts at **(C, D)** and ends at **(423, 428)**. If we draw this problem using the **DDA algorithm**, determine the number of times **x** will be increased. How many pixels will be needed to draw this line?
$$x/7 + y/12 = 5$$
6. Suppose a line segment has the equation above, and the starting point is at **y = D** and the ending point is at **y = 0**. Find the **first 6 pixels** using the Midpoint Line Algorithm.

7. A clipping window is **C** units wide and **A** units tall, and has its center at **(B, D)**. The endpoints of a line segment are **(-45, 65)** and **(23, -10)**. Apply **Cohen-Sutherland Algorithm**, and identify whether the line is "**partially inside**", "**completely inside**", or "**completely outside**". If it is the first case, run the algorithm to calculate **new endpoints** for the line segment so that it is inside the clipping window. Does **Cyrus-Beck line clipping algorithm** produce the same output?
8. Arya drew a triangle on a coordinate plane with vertices **A(3, 2)**, **B(5, 1)** and **C(4, 3)**. She performed the following transformations in sequence: At first, she **translated** the triangle by **(-2, 3)**. Then, **rotate** the triangle **90° counterclockwise** about the point **(C, D)**. After that, she **reflected** the triangle on the **x-axis**. Finally, she **scaled** the triangle **uniformly** by a factor of **2** about the point **(1, -1)**. After applying these transformations, the new location of one vertex was found to be (a, b).
- Now, identify the position of the vertices **before only the reflection is applied**. (That means only translation and rotation are applied to each of the vertices).
 - Write the **composite matrix formulation** for all transformations applied to the triangle. (You do not need to perform matrix multiplication.)
 - Which **geometric properties are preserved** after each transformation?
9. A **3D composite transformation** is defined by a **shearing** along the **X axis** by a shearing factor of **(2, 4)** about point **(423, 0, 0)**, followed by a **uniform scaling** by **factor 3** again, followed by a **30-degree counterclockwise** rotation on the **X axis** about point **(2, 10, 12)** and finally followed by a **translation** of **(-4, -2, -3)**. A point **P** is transformed into **P'** with the above transformation. Now, write down the **composite transformation matrix** representation for **P'** in the correct sequence of matrix multiplications. [N.B. The shearing factor will be **Sh_y = 2** and **Sh_z = 4**. Go sequentially if it is not mentioned.]
10. A **composite transformation** is defined by a scaling on the **X-axis** and **Y-axis** (first transformation) followed by a second transformation, via the following equations:

$$x' = 5x - 11$$

$$y' = 10y + 22$$

$$z' = 33 + z$$

- Find the **composite transformation matrix**. What was the **second transformation** after scaling?
- Write down the **scaling matrix** from the above composite transformation.
- If a **3D** point **M** has the coordinate **(B, C, D)**, what is the new coordinate of the point **M'**, after the transformation defined by the equations?

Submission Guidelines:

- You MUST write the answers on the paper **by hand**. You can also use a Tab.
- Make a cover page including your **Assignment Number, Name, ID, Section, and Date**.
- Scan the pages and make a PDF. The file name should follow the format: StudentID_Section_Assignment_1 (eg **10301040_6_Assignment_1**)
- **No late submissions will be accepted.**
- Upload the PDF to the Google form.